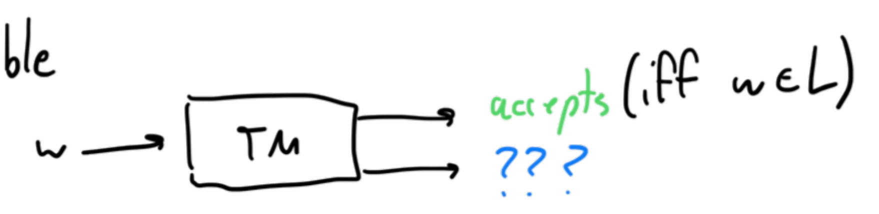
Ch. 4 Decidable Problems

Decidable

Recognizable



Does a given DFA accept a given string?
The following language contains all encodings of DFA's with their accepted strings:

$$\boxed{A_{DFA} = \{\langle B, w\rangle \mid B \text{ is a DFA that accepts string } w\}}$$

The problem of testing whether B accepts w is the same as testing whether $\langle B, w\rangle \in A_{DFA}$.

Theorem 4.1 A_{DFA} is a decidable language.

Proof: We present a TM M that decides A_{DFA}.
M = " On input $\langle B, w\rangle$, where B is a DFA and w is a string:
1. Simulate B on input w.
   we can hold the current state of B on the tape. The tape keeps track of everything.
2. If the simulations ends in an accept state, accept. Otherwise, it ends in a nonaccepting state, reject."

$\square$

$$A_{NFA} = \{\langle B, w\rangle \mid B \text{ is an NFA that accepts string } w\}.$$

Theorem 4.2 A_{NFA} is a decidable language.

N = " On input $\langle B, w\rangle$, where B is an NFA and w is a string:
1. Convert NFA B into an equivalent DFA C using subset substition (Thm 1.39).
2. Run TM M from Theorem 4.1 on input $\langle C, w\rangle$.
3. If M accepts, accept. Otherwise, reject." $\square$

Theorem 4.3 Let $A_{REX} = \{\langle R, w\rangle \mid R \text{ is a regex that generates string } w\}$

P = " On input $\langle R, w\rangle$, where R is regex and w is a string:
1. Convert R to an NFA A using conversion given by Thm 1.54.
2. Run TM N on $\langle A, w\rangle$ from Thm 4.2.
3. If N accepts, accept. Otherwise, reject."

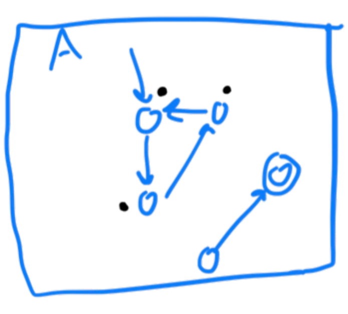
$\square$

Theorem = major result
Lemma = a "function" in a theorem
Corollary = a result implied by a Theorem
Proposition = minor results not to be used a lemma, generally.

$$E_{DFA} = \{\langle A\rangle \mid A \text{ is a DFA and } L(A) = \varnothing\}$$

Theorem 4.4 E_{DFA} is decidable.

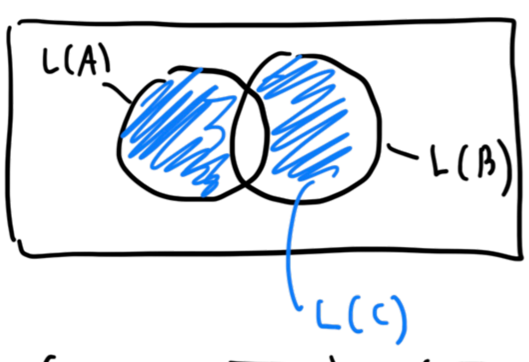
T = " On input $\langle A\rangle$, where A is a DFA:
1. Mark the start state of A.
2. Repeat 3 until no new states get marked.
3. Mark any state that has a transition from a marked state.
4. If no accept state is marked, accept. Otherwise, reject." $\square$



$$EQ_{DFA} = \{\langle A, B\rangle \mid A \text{ \& } B \text{ are DFA's and } L(A) = L(B)\}.$$

Theorem 4.5 EQ_{DFA} is decidable.

Proof: We will construct a DFA C from A & B, where C accepts only strings accepted by one but not both of $\{A, B\}$.



symmetric difference.

$$L(C) = \left(L(A) \cap \overline{L(B)}\right) \cup \left(\overline{L(A)} \cap L(B)\right)$$

F = " On input $\langle A, B\rangle$, where A & B are DFA's:
1. Construct DFA C as described.
2. Run TM T from Thm 4.4 on input $\langle C\rangle$.
3. If accepts, accept. Otherwise, reject." $\square$